
MAIAEDGE NETWORKS • ENGINEERING

Statement of Work

Nexus Multi-Building Connectivity Architecture

Engineering Phase

DOCUMENT VERSION	v1.0 — Draft for Review
PREPARED BY	MaiaEdge Networks, Inc.
PREPARED FOR	Nexus Data Centers
DATE	May 2026
CLASSIFICATION	Confidential & Proprietary
DOCUMENT OWNER	MaiaEdge Engineering Lead

SECTION 01**OBJECTIVE**

This Statement of Work defines the engineering scope required to de-risk the delivery schedule for the initial 4-building deployment at Nexus Data Centers, while establishing scalable architecture capable of supporting up to 65 Nexus data center buildings across the portfolio.

This engineering phase is structured to achieve three primary outcomes:

- Achieve approximately 60% engineering completion across all design domains
- Produce procurement-ready outputs including full Bill of Materials and vendor specifications
- Enable a seamless transition to construction and deployment execution phases

This SOW covers engineering services only. Procurement execution, physical installation, and construction activities are explicitly excluded unless separately scoped.

SECTION 02**SCOPE OF WORK**

2.1 Phase 1 — Initial 4-Building Deployment

MaiaEdge will develop a Low-Level Design (LLD) package covering all infrastructure layers required for the initial deployment:

- Meet-Me Room (MMR) / Carrier Hotel architecture and cage layout
- MaiaEdge PBC hardware deployment — placement, mounting, power, and cabling
- Fiber ingress and building entry point design
- Distribution layer topology and interconnect model
- Switch fabric integration — architecture and uplink design

END-TO-END CONNECTIVITY MODEL

Carrier → MMR → PBC → Dark Fiber → Building → PBC → Switch Fabric

2.2 Fiber & Infrastructure Design

Requirements definition will cover:

- Dark fiber ingress into the MMR and associated ODF cross-connect model
- Diverse fiber paths (A/B routes) to each building for high-availability design
- Interbuilding dark fiber mesh topology

Design deliverables will include:

- Building-to-building connectivity model (partial or full mesh, based on cost/scale trade-off)
- Fiber path diversity and routing assumptions documentation
- Defined handoff points between fiber provider, MMR, and each Nexus building

2.3 MaiaEdge Infrastructure Design

MaiaEdge will define the hardware architecture across all sites:

- Dual cage architecture within the MMR — Primary and Secondary
- Dual PBC deployment per cage (PBC 2000 series) — Primary / Secondary failover model
- Symmetric connectivity from all PBCs to buildings — no single points of failure
- PBC placement, power, and connectivity design within each Nexus building
- Integration specifications between PBC and switching fabric

2.4 Expansion Architecture (Up to 65 Buildings)

A scalable network CPE architecture framework will be developed in parallel with Phase 1 design, covering:

- Fiber expansion model — incremental fiber provisioning as new buildings are added
- Mesh and interconnect strategy — partial to full mesh transition thresholds
- Capacity planning assumptions — bandwidth, port density, and PBC scaling
- Modular deployment approach — standardized build-out templates per building

2.5 Interface Coordination

Coordinate across all infrastructure interface parties:

- Dark fiber provider — critical dependency; will drive off-site routing and on-site delivery schedule
- Carrier coordination — based on current engagement model; inputs required for design assumptions
- Installation partners — constructability feedback and duration estimate inputs

Coordination activities include: regular design interface meetings, cross-party alignment sessions, and interface block diagram maintenance.

2.6 Owner-Provided or Builder Data Required Information

This information is crucial so Pearce / CBRE technicians can install the equipment based upon design requirements.

- OSP Engineering Data
 - › Geospatial OSP fiber routing plans aligned to site layout
 - › Conduit pathway requirements and alignment
 - › Handhole / vault placement recommendations
 - › Building entry location definition
 - › Cable routing and segmentation plans
 - › Cable sizing and allocation (strand-level backbone definition)
 - › Splice enclosure locations and design
 - › Preliminary splice matrices
 - › Diversity validation (A/B physical separation)
- ISP Engineering Data
 - › Telecom room layouts (MMR, BDF, MDF, IDF as applicable)
 - › ODF layout and rack elevations
 - › Fiber pathway design within buildings (tray / riser coordination)
 - › Entrance → distribution room routing plans
 - › A/B pathway separation validation within building
 - › Data hall fiber distribution design (aligned to actual hall layouts)
 - › End-of-row / top-of-rack distribution layouts
 - › Tenant-ready distribution framework (if applicable)

- **Fiber Systems Engineering & Documentation**
 - › Strand-level fiber assignments
 - › Final splice matrices
 - › ODF port assignments
 - › Patch panel layouts
 - › Cross-connect mapping
 - › Pre-terminated trunk cable schedules
 - › Labeling schema implementation (input may be required from Nexus team)
 - › Fiber naming conventions, finalized (input may be required from Nexus team)
 - › Documentation structure for fiber management system (input may be required from Nexus team)
- **Construction Coordination & Technical Oversight**
 - › Participation in coordination meetings (GC, civil, MEP trades, low-voltage contractors)
 - › Review of contractor submittals
 - › Validation of installation against design intent
 - › Field issue resolution support
 - › Design clarification responses
 - › Punch list support
 - › Readiness reviews prior to testing / turnover
- **Site & Building Data**
 - › Site master plan (CAD / BIM)
 - › Building floorplans
 - › Telecom Hotel — room and / or cage locations and sizes
 - › Data Halls — room and / or cage locations and sizes

SECTION 03

DELIVERABLES & DELIVERY SCHEDULE

All deliverables are structured against a T+0 engagement start date. The table below defines each deliverable, its category, and its target delivery milestone.

DELIVERABLE	CATEGORY	TARGET
Kickoff Meeting — Agenda, Attendees, Decision Log	Project Initiation	T+0 (Day 1)
Technical Input Request — Required data from Nexus / fiber provider	Project Initiation	T+0 (Day 1)
Baseline Architecture Assumptions Document	Design Foundation	T+2 weeks
Interface Block Diagram (Fiber Provider / Carrier / Nexus / MaiaEdge)	Design Foundation	T+2 weeks
Phase 1 Low-Level Design (LLD) — All 4 buildings	Phase 1 Design	T+4 weeks
MMR / Carrier Hotel Architecture Design Package	Phase 1 Design	T+4 weeks

Building Connectivity Design — Per-building topology	Phase 1 Design	T+4 weeks
Interbuilding Fiber Mesh Design	Phase 1 Design	T+4 weeks
Bill of Materials (BOM) — Phase 1, equipment & fiber	Procurement Readiness	T+6 weeks
Material Takeoff (MTO) — Structured by building and layer	Procurement Readiness	T+6 weeks
Long-Lead Item Identification & Vendor Recommendation	Procurement Readiness	T+6 weeks
Building Infrastructure Specifications	Procurement Readiness	T+6 weeks
Fiber Design & Routing Assumptions Document	Procurement Readiness	T+6 weeks
MaiaEdge PBC Deployment Specifications (all sites)	Procurement Readiness	T+6 weeks
Vendor-Neutral Equipment Specifications & Data Sheets	Procurement Readiness	T+6 weeks
Expansion Architecture Document (to 65 buildings)	Expansion Planning	T+8 weeks
Capacity Planning Model — Bandwidth, port density, PBC scale	Expansion Planning	T+8 weeks
Construction-Ready Handoff Package — Phase 1	Execution Prep	T+12 weeks
Installation Duration Estimates (per building)	Execution Prep	T+12 weeks
High-Level Cost Model — Infrastructure, Equipment, Deployment	Execution Prep	T+12 weeks
GMP / NTE Readiness Package	Commercial Enablement	T+12 weeks

SECTION 04

PROJECT MILESTONES

MILESTONE	DELIVERABLE GATE	TARGET WEEK
M1 — Engagement Launch	Technical inputs received; kickoff complete; assumptions baseline published	Week 1
M2 — Architecture Foundation	Interface Block Diagram approved; LLD scope locked	Week 2
M3 — Phase 1 LLD Complete	All 4-building designs complete and under review	Week 4
M4 — Procurement Package Complete	BOM, MTO, specs, and long-lead items delivered	Week 6–8
M5 — Expansion Design Complete	65-building architecture and capacity model delivered	Week 8
M6 — Construction Handoff (~60%	Construction-ready package, cost model, GMP	Week 12

Engineering)

readiness

SECTION 05

ROLES & RESPONSIBILITIES

5.1 Team Structure

ROLE	INDIVIDUAL / PARTY	RESPONSIBILITIES
Engagement Lead	MaiaEdge — Senior Network Architect	Overall accountability, commercial alignment, escalation point, Nexus executive relationship
Engineering Lead	MaiaEdge — Senior Network Architect	Technical design ownership, LLD delivery, peer review coordination
PBC Integration Specialist	MaiaEdge — Hardware Engineering	PBC deployment specs, hardware design, integration with switching fabric
Fiber Design Lead	MaiaEdge / Third-Party PE	Dark fiber design, path diversity, interbuilding mesh, ODF cross-connect model
Project Coordinator	MaiaEdge	Schedule tracking, milestone management, interface meeting coordination, document control
Nexus Technical POC	Nexus Data Centers	Provide site requirements, review deliverables, approve design assumptions, interface with fiber provider
Dark Fiber Provider	TBD	Off-site fiber routing, on-site delivery, handoff point coordination
Installation Partner	TBD	Constructability input, duration estimates, site survey coordination

SECTION 06

INTERFACES & DEPENDENCIES

INTERFACE PARTY	DEPENDENCY	CRITICALITY	REQUIRED BY
Dark Fiber Provider	Off-site routing plan, on-site fiber delivery schedule, handoff point confirmation	Critical	T+1 week
Nexus Data Centers	Initial building list (4 sites), MMR requirements, client-specific requirements	Critical	T+0

Carrier / Third Party	Carrier coordination status; assumptions for design basis	High	T+2 weeks
Installation Partner	Constructability input, site access assumptions, duration estimates	Medium	T+4 weeks

SECTION 07

DESIGN BASIS & ASSUMPTIONS

- Standardized, non-bespoke architecture applied uniformly across all 4 Phase 1 buildings
- Dual-path (A/B) dark fiber design for high availability at all building entry points
- Symmetric PBC 2000 architecture deployed consistently across MMR cages and all buildings
- Partial or full mesh interbuilding connectivity — final topology selected based on cost / scale analysis
- Carrier coordination is managed by an existing third party; inputs will be provided as-available
- Client-specific requirements (if any) are assumed at high level and may evolve during engagement
- All design outputs are vendor-neutral unless otherwise specified by Nexus
- Phase 1 buildings and MMR location are assumed to be within the Hubbard, TX Nexus campus

SECTION 08

TECHNICAL INPUTS REQUIRED AT T+0

The following inputs are required from Nexus Data Centers and associated parties at or before engagement commencement to avoid schedule impact:

INPUT	OWNER	REQUIRED BY
Confirmed list of 4 Phase 1 buildings (addresses / building IDs)	Nexus	T+0
MMR location, dimensions, and power / cooling constraints	Nexus	T+0
Dark fiber routes and proposed handoff points	Fiber Provider	T+1 week
Carrier coordination status and point of contact	Nexus / Third Party	T+1 week
Any Nexus client-specific connectivity requirements	Nexus	T+2 weeks
Installation contractor point of contact	Nexus	T+3 weeks
Existing infrastructure drawings (if any)	Nexus	T+0

SECTION 09

EXCLUSIONS

The following are explicitly out of scope for this Statement of Work:

- Procurement execution — placement of purchase orders or vendor contracts
- Physical installation, construction, or site commissioning activities
- Carrier-specific custom design work (unless separately scoped and agreed)
- Operational or managed service design
- Network management, monitoring platform, or NOC integration design
- Any design activity dependent on inputs not received by the required dates above

SECTION 10

COMMERCIAL BASIS

This engineering phase is structured to enable a predictable commercial framework for subsequent phases:

- Deliverables will provide sufficient design maturity to support Guaranteed Maximum Price (GMP) discussions
- Engagement may be structured on a Not-to-Exceed (NTE) commercial basis
- Phase 1 engineering completion (~60%) is designed to de-risk procurement and installation pricing
- Expansion architecture work (Section 2.4) enables long-range portfolio planning and phased investment commitments

A detailed commercial proposal inclusive of fees, payment milestones, and T&M provisions will be provided under separate cover.

SECTION 11

ACCEPTANCE & AUTHORIZATION

This Statement of Work becomes effective upon execution by authorized representatives of both parties. By signing below, each party confirms they have read, understood, and agreed to the terms set forth herein.

MAIAEDGE NETWORKS, INC.	
AUTHORIZED SIGNATORY	
NAME	
TITLE	
DATE	

NEXUS DATA CENTERS	
AUTHORIZED SIGNATORY	
NAME	
TITLE	
DATE	

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